Lab Sheet 1
Database Management System Lab (BTCS203P)
B. TECH AI&ML, CYBER SECURITY, Data Science 3 RD Semester (Session 2026-2025)

Objective:
To understand and practice Data Definition Language (DDL) commands for creating and modifying the structure of a table in a relational database using SQL (MySQL).
Prerequisites:
1. Basic knowledge of SQL.
2. MySQL installed and set up.

Task : Open SQL client and perform the following tasks
Task 1: Create a Table
a. Create a database named Company.
= CREATE DATABASE Company;
b. Create a table named Employees with the following structure:

Column Name	Data Type	Constraints
EmpID AUTO_INCREMENT	INT	PRIMARY KEY
FirstName	VARCHAR(50)	NOT NULL
LastName	VARCHAR(50)	
Department	VARCHAR(50)	
Salary	DECIMAL(10,2)	
Email	VARCHAR(100)	UNIQUE

```
=USE Company;
CREATE TABLE Employees (
   EmployeeID INT AUTO_INCREMENT PRIMARY KEY,
   FirstName VARCHAR(50) NOT NULL,
   LastName VARCHAR(50),
   Department VARCHAR(50),
   Salary DECIMAL(10,2),
   Email VARCHAR(100) UNIQUE
);
```

Task 2: DDL Queries on the Employees Table
a) Describe the structure of the table.

=DESCRIBE Employees;
b) Rename the table to Staff.
=RENAME TABLE Employees TO Staff;

c)Add a new column PhoneNumber (VARCHAR(15)).
=ALTER TABLE Staff
ADD PhoneNumber VARCHAR(15);

d) Modify the data type of Salary to DECIMAL(12,2).
=ALTER TABLE Staff
MODIFY LastName VARCHAR(50);

e) Add a DEFAULT value of 'General' to the Department column.
=ALTER TABLE Staff
ALTER COLUMN Department SET DEFAULT 'General';

f) Drop the Phone Number column.
=ALTER TABLE Staff
DROP COLUMN PhoneNumber;

g) Add a CHECK constraint to ensure Salary is greater than 0.
=ALTER TABLE Staff
ADD CONSTRAINT chk_Salary CHECK (Salary > 0);

h) Rename the column First Name to Given Name.
=ALTER TABLE Staff
RENAME COLUMN FirstName TO GivenName;

i) Drop the chk_Salary constraint:
=ALTER TABLE Staff
DROP CHECK chk_Salary;

j) Create an index on the Email column.
=CREATE INDEX idx_email
ON Staff(Email);

k) Drop the index on the Email column.
=DROP INDEX idx_email ON Staff;

l) Truncate the table to remove all rows without affecting the structure.
=TRUNCATE TABLE Staff;

m) Drop the Staff table completely
=DROP TABLE Staff;

<u>3. Conceptual Questions:</u>

1. What is the difference between TRUNCATE and DELETE?
=**TRUNCATE** removes **all records** from a table at once, while **DELETE** can remove specific records using a WHERE condition.

- TRUNCATE cannot normally use WHERE.
- DELETE can use WHERE.
- TRUNCATE is generally faster for removing all rows.
- Both keep the table structure.

Example:

TRUNCATE TABLE Employees;

DELETE FROM EmployeesWHERE EmployeeID = 5;

2. Explain the use of the CHECK constraint.

=The **CHECK constraint** is used to ensure that the values entered into a column satisfy a specific condition.For example:

Salary DECIMAL(10,2),CHECK (Salary > 0)

This ensures that the salary cannot be **zero or negative**.It helps maintain **valid and accurate data** in the table.

3. Why is it important to define a PRIMARY KEY for a table?
=A **PRIMARY KEY** uniquely identifies each record in a table.

For example:

Employee ID INT PRIMARY KE

Importance:

- It prevents duplicate values.
- It does not allow NULL values.
- It uniquely identifies each row.
- It helps maintain **data integrity**.
- It makes it easier to reference a particular record.

4. How does ALTER TABLE help in modifying an existing table?
=ALTER TABLE is used to **change the structure of an existing table** without creating a new table.
· Add a new column.
· Delete a column.

· Modify a column's data type.
· Rename a column.
· Add or remove constraints.

EXAMPLE:
ALTER TABLE EmployeesADD PhoneNumber VARCHAR(15);
This adds a new Phone Number column to the existing table.

5. What is the purpose of creating an index in a table?
=An **index** is created to make searching and retrieving data from a table **faster**.
EXAMPLE:
CREATE INDEX idx_email ON Employees(Email);

This creates an index on the Email column, which can improve the speed of queries that search using that column.

4. Exercises:

1. Create a table named Projects with the following structure:

a) Perform at least 5 DDL operations on the Projects table.

Column Name	Data Type	Constraints
ProjectID	INT	PRIMARY KEY AUTO_INCREMENT
ProjectName	VARCHAR(100)	NOT NULL
StartDate	DATE	
EndDate	DATE	
Budget	DECIMAL(15,2)	CHECK (Budget > 0)

```
=CREATE TABLE Projects (
   ProjectID INT AUTO_INCREMENT PRIMARY KEY,
   ProjectName VARCHAR(100) NOT NULL,
   StartDate DATE,
   EndDate DATE,
   Budget DECIMAL(15,2),
   CONSTRAINT chk_Budget CHECK (Budget > 0)
);
DESCRIBE projects;
//alter table
ALTER TABLE Projects
//add column
ADD ManagerName VARCHAR(100);
//Modify a column
ALTER TABLE Projects MODIFY Description VARCHAR(500);
//Rename a column
ALTER TABLE Projects RENAME COLUMN Status TO ProjectStatus;
//Drop a column
ALTER TABLE Projects DROP COLUMN ProjectStatus;
```

b) Write a query to retrieve the names of all columns in the Staff table.
=SELECT COLUMN_NAME
FROM INFORMATION_SCHEMA.COLUMNS
WHERE TABLE_SCHEMA = DATABASE()
AND TABLE_NAME = 'Staff';
DESCRIBE Staff;

c) Write a query to modify the Department column to allow NULL values.
=Department VARCHAR(50) NOT NULL
ALTER TABLE Staff
MODIFY Department VARCHAR(50) NULL;

Q1 RESULTS

```
DROP DATABASE IF EXISTS Company;
CREATE DATABASE Company;
USE Company;
CREATE TABLE Employee (ID INT AUTO_INCREMENT PRIMARY KEY, FirstName VARCHAR(50) NOT NULL,
    LastName VARCHAR(50),
    Department VARCHAR(50),
    Salary DECIMAL(10,2),
    Email VARCHAR(100) UNIQUE);
SHOW TABLES;
DESCRIBE Employee;
Insert INTO Employee( FirstName, LastName, Department,Salary,Email )
VALUES
("Suhani","Saini", "Btech",60000, "Suhani@1234");
```

esult Grid | Filter Rows: | Export: | Wrap Cell Content:

Field	Type	Null	Key	Default	Extra
ID	int	NO	PRI	NULL	auto_increment
FirstName	varchar(50)	NO		NULL	
LastName	varchar(50)	YES		NULL	
Department	varchar(50)	YES		NULL	
Salary	decimal(10,2)	YES		NULL	

Q4

Query 1 | SQL File 3* | SQL File 4* | SQL File 5*

Limit to 1000 rows

```
DROP TABLE IF EXISTS Projects;
CREATE TABLE Projects (
    ProjectID INT AUTO_INCREMENT PRIMARY KEY,
    ProjectName VARCHAR(100) NOT NULL,
    StartDate DATE,
    EndDate DATE,
    Budget DECIMAL(15,2),
    CONSTRAINT chk_Budget CHECK (Budget > 0)
);
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Field	Type	Null	Key	Default	Extra
ProjectID	int	NO	PRI	NULL	auto_increment
ProjectName	varchar(100)	NO		NULL	
StartDate	date	YES		NULL	
EndDate	date	YES		NULL	
Budget	decimal(15,2)	YES		NULL	

```sql
ALTER TABLE Projects
ADD Status VARCHAR(20);
ALTER TABLE Projects
MODIFY Description VARCHAR(500);
ALTER TABLE Projects
RENAME COLUMN Status TO ProjectStatus;
ALTER TABLE Projects
DROP COLUMN ProjectStatus;
DESCRIBE Projects;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Field	Type	Null	Key	Default	Extra
ProjectID	int	NO	PRI	NULL	
ProjectName	varchar(100)	YES		NULL	
Description	varchar(500)	YES		NULL	
StartDate	date	YES		NULL	
EndDate	date	YES		NULL	

Q4

Q4 b

```sql
ALTER TABLE Staff
MODIFY Salary DECIMAL(12,2);
ALTER TABLE Staff MODIFY Department VARCHAR(50) DEFAULT 'GENERAl';
INSERT INTO Staff(FirstName, LastName,Salary,Email )
VALUES('SONAM','RAJ',50000,'SO1234@23');
SHOW TABLEs;
DESC Staff;
INSERT INTO Staff (Firstname, LastName, Department, Salary, Email)
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

ID	FirstName	LastName	Department	Salary	Email
1	Suhani	Saini	Btech	60000.00	Suhani@1234
NULL	NULL	NULL	NULL	NULL	NULL